

Typesetting captions with the caption package*

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Abstract

The caption package offers customization of captions in floating environments such `figure` and `table` and cooperates with many other packages.¹

Contents

1	Introduction	2
2	Using the package	3
3	Options	3
3.1	Formatting	3
3.2	Justification	6
3.3	Fonts	7
3.4	Margins and further paragraph options	8
3.5	Styles	10
3.6	Skips	10
4	Useful stuff	11
5	Do it yourself!	14
5.1	Further Examples	16
6	Using non-standard document classes	17
7	Compatibility to other packages	18
7.1	The float package	18
7.2	The listings package	19
7.3	The longtable package	19

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¹A complete re-work of the user interface done with Steven D. Cochran and Frank Mittelbach has lead to this new enhanced version 3.0.

7.4	The rotating package	20
7.5	The sidecap package	20
7.6	The supertabular package	20
7.7	Known incompatibilities	20
8	Compatibility to older versions	20
8.1	The caption package version 1.x	20
8.2	The caption2 package version 2.x	22
9	Further reading	24
10	Thanks	24

1 Introduction

Within the standard L^AT_EX classes captions haven't received the attention they deserve. Simply typeset as an ordinary paragraph there is no remarkable visual difference from the rest of the text, like here:

Figure 1: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

There should be possibilities to change this; for example, it would be nice if you could make the text of the caption a little bit smaller as the normal text, add an extra margin, typeset the caption label with the same font family and shape as your headings etc. Just like this one:

Figure 2 – White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

You can do this easily with this package as there are many ready-to-use caption formatting options, but you are free to define your very own stuff, too.

2 Using the package

`\usepackage` Insert

```
\usepackage[\langle options \rangle]{caption}[2007/04/09]
```

into the preamble of your document, i.e. the part of your document between `\documentclass` and `\begin{document}`. The options control how your captions will look like; e.g.,

```
\usepackage[margin=10pt,font=small,labelfont=bf]{caption}
```

would result in captions looking like the second one in the introduction.

`\captionsetup` For a later change of options the caption package provides the command

```
\captionsetup[\langle float type \rangle]{\langle options \rangle}
```

So

```
\usepackage[margin=10pt,font=small,labelfont=bf]{caption}
```

and

```
\usepackage{caption}
\captionsetup{margin=10pt,font=small,labelfont=bf}
```

are equal in their results.

It's good to know that `\captionsetup` has an effect on the current environment only. So if you want to change some settings for the current figure or table only, just place the `\captionsetup` command inside the figure or table right before the `\caption` command. For example

```
\begin{figure}
...
\captionsetup{singlelinecheck=off}
\caption{...}
\end{figure}
```

switches the single-line-check off, but only for this figure so all the other captions remain untouched.

(For a description of the optional parameter *\langle float type \rangle* see section 4: “Useful stuff”.)

3 Options

3.1 Formatting

`format=` A figure or table caption mainly consists of three parts: the caption label, which says if this object is a ‘Figure’ or ‘Table’ and what number is associated with it, the caption text itself, which is normally a short description of contents, and the caption separator which separates the text from the label.

The *caption format* determines how this information will be presented; it is specified with the option

`format=\langle format name \rangle` ,

having the name of the caption format as its argument.

There are two standard caption formats:

New description v3.0h	<code>plain</code>	Typesets the captions as a normal paragraph. (This is the default behaviour, it is adapted from the standard $\text{\LaTeX}$ document classes.)
	<code>hang</code>	Indents the caption text, so it will ‘hang’ under the first line of the text.
	...	Own formats can be defined using <code>\DeclareCaptionFormat</code> . (See section 5: “ <i>Do it yourself</i> ”)

An example: Specifying the option

`format=hang`

yields captions like this:

Figure 3: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`indentation=` For both formats (`plain` and `hang`) you can setup an extra indentation starting at the second line of the caption. You do this with the option

`indentation=\langle amount \rangle`.

Two examples:

`format=plain,indentation=.5cm`

Figure 4: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`format=hang,indentation=-0.5cm`

Figure 5: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`labelformat=` With the option

`labelformat=\langle label format name \rangle`

New description
v3.0e you specify how the caption label will be typeset. There are four standard caption label formats:

`default` The caption label will be typeset as specified by the document class, usually this means the name and the number (like `simple`). (This is the default behaviour.)

<code>empty</code>	The caption label will be empty. (This option makes sense when used together with other options like <code>labelsep=none</code> .)
<code>simple</code>	The caption label will be typeset as a name and a number.
<code>parens</code>	The number of the caption label will be typeset in parentheses.
<code>...</code>	Own label formats can be defined using <code>\DeclareCaptionLabelFormat</code> . (See section 5: “ <i>Do it yourself</i> ”)

An example: Using the options

```
labelformat=parens, labelsep=quad
```

gives captions like this one:

Figure (6) White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`labelsep=` With the option

```
labelsep=<label separator name>
```

you specify what caption separator will be used. You can choose one of the following:

<code>none</code>	There is no caption separator. (This option makes sense when used together with other options like <code>labelformat=empty</code> .)
<code>colon</code>	The caption label and text will be separated by a colon and a space. (This is the default one.)
<code>period</code>	The caption label and text will be separated by a period and a space.
<code>space</code>	The caption label and text will be separated by a single space.
<code>quad</code>	The caption label and text will be separated by a <code>\quad</code> .
<code>newline</code>	The caption label and text will be separated by a line break (<code>\\</code>).
New feature v3.0h <code>endash</code>	The caption label and text will be separated by an en-dash, surrounded by spaces (<code>--</code>).
<code>...</code>	Own separators can be defined using <code>\DeclareCaptionLabelSeparator</code> . (See section 5: “ <i>Do it yourself</i> ”)

Three examples:

```
labelsep=period
```

Figure 7. White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

```
labelsep=newline,singlelinecheck=false
```

Figure 8

White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

```
labelsep=endash
```

Figure 9 – White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

3.2 Justification

`justification=` As addition to the caption format you could also specify a *caption justification*; it is specified with the option

```
justification=<justification name> .
```

You can choose one of the following:

<code>justified</code>	Typesets the caption as a normal paragraph. (This is the default.)
<code>centering</code>	Each line of the caption will be centered.
<code>centerlast</code>	The last line of each paragraph of the caption text will be centered.
<code>centerfirst</code>	Only the first line of the caption will be centered.
<code>raggedright</code>	Each line of the caption will be moved to the left margin.
<code>RaggedRight</code>	Each line of the caption will be moved to the left margin, too. But this time the command <code>\RaggedRight</code> of the <code>ragged2e</code> package will be used to achieve this. The main difference is that the word breaking algorithm of <code>T_EX</code> will work inside captions.
<code>raggedleft</code>	Each line of the caption will be moved to the right margin.
<code>...</code>	Own justifications can be defined using <code>\DeclareCaptionJustification</code> . (See section 5: “ <i>Do it yourself</i> ”)

Three examples:

```
justification=centerlast
```

Figure 10: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

```
format=hang, justification=raggedright
```

Figure 11: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

```
labelsep=newline, justification=centering
```

Figure 12

White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`singlelinecheck=` The standard L^AT_EX document classes (`article`, `report`, and `book`) automatically center a caption if it fits in one single line:

Figure 13: A short caption.



The caption package adapts this behaviour and therefore usually ignores the justification you have set with `justification=` in such case. But you can switch this special treatment of such short captions off with the option

```
singlelinecheck=<bool> .
```

Using `false`, `no`, `off` or `0` for *<bool>* switches the extra centering off:

```
singlelinecheck=false
```

Doing so the above short caption would look like

Figure 13: A short caption.

You switch the extra centering on again by using `true`, `yes`, `on` or `1` for *<bool>*. (The default is on.)

3.3 Fonts

`font=` There are three font options which affects different parts of the caption: One affecting the whole caption (`font`), one which only affects the caption label and separator (`labelfont`) and at least one which only affects the caption text (`textfont`). You set them up using the options

```
font={ <font options> } ,
labelfont={ <font options> } , and
textfont={ <font options> } .
```

And these are the available font options:

```
scriptsize    Very small size
```

<code>footnotesize</code>	The size usually used for footnotes
<code>small</code>	Small size
<code>normalsize</code>	Normal size
<code>large</code>	Large size
<code>Large</code>	Even larger size
<code>up</code>	Upright shape
<code>it</code>	<i>Italic shape</i>
<code>sl</code>	<i>Slanted shape</i>
<code>sc</code>	SMALL CAPS SHAPE
<code>md</code>	Medium series
<code>bf</code>	Bold series
<code>rm</code>	Roman family
<code>sf</code>	Sans Serif family
<code>tt</code>	Typewriter family
<code>...</code>	Own font options can be defined using <code>\DeclareCaptionFont</code> . (See section 5: “Do it yourself”)

If you use only one of these options you can omit the braces; e.g., the options `font={small}` and `font=small` will give the same result.

Two examples:

```
font={small,it},labelfont=bf
```

Figure 14: *White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.*

```
font=small,labelfont=bf,textfont=it
```

Figure 15: *White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.*

3.4 Margins and further paragraph options

`margin=` For all captions you can specify *either* an extra margin *or* a fixed width. You do this by
`width=` using the options

```
margin=<amount> or
width=<amount>
```

Nevertheless what option you use, the left and right margin will be the same.

Two examples will illustrating this:

```
margin=10pt
```

Figure 16: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

```
width=.75\textwidth
```

Figure 17: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`parskip=` This option is useful for captions containing more than one paragraph. It specifies the extra vertical space inserted between them:

```
parskip=<amount>
```

One example:

```
margin=10pt,parskip=5pt
```

Figure 18: First paragraph of the caption. This one contains some test, just to show how these options affect the layout of the caption.

Second paragraph of the caption. This one contains some text, too, to show how these options affect the layout of the caption.

`hangindent=` The option

```
hangindent=<amount>
```

is for setting up a hanging indentation starting from the second line of each paragraph. If the caption contains just a single paragraph, using this option leads to the same result as the option `indentation=` you already know about. But if the caption contains multiple paragraphs you will notice the difference:

```
format=hang,indentation=-.5cm
```

Figure 19: First paragraph of the caption. This one contains some test, just to show how these options affect the layout of the caption.

Second paragraph of the caption. This one contains some text, too, to show how these options affect the layout of the caption.

```
format=hang,hangindent=-.5cm
```

Figure 20: First paragraph of the caption. This one contains some test, just to show how these options affect the layout of the caption.

Second paragraph of the caption. This one contains some text, too, to show how these options affect the layout of the caption.

3.5 Styles

`style=` A suitable combination of caption options is called *caption style*. You can compare them more or less to page styles which you set up with `\pagestyle`: The caption style provides all settings for a whole caption layout.

You switch to an already defined caption style with the option

```
style=<style name> .
```

The caption package usually defines only the style `default` which puts all options you already know about to the default ones. This means that specifying the option

```
style=default
```

has the same effect as specifying all these options:

```
format=default, labelformat=default, labelsep=default,
justification=default, font=default, labelfont=default,
textfont=default, margin=0pt, indentation=0pt, parindent=0pt
hangindent=0pt, singlelinecheck=true
```

Own caption styles can be defined using `\DeclareCaptionStyle`. (See section 5: “Do it yourself”)

3.6 Skips

`aboveskip=` The spaces above and below the caption are controlled by the skips `\abovecaptionskip`
`belowskip=` and `\belowcaptionskip`. The standard L^AT_EX document classes `article`, `report` and `book` set `\abovecaptionskip` to 10pt and `\belowcaptionskip` to 0pt. Both skips can be changed with the command `\setlength`, but you can use these options, too:

```
aboveskip=<amount> and
belowskip=<amount> .
```

`position=` Using `\abovecaptionskip` and `\belowcaptionskip` has a major design flaw: If the caption is typeset *above* (and not *below*) the figure or table they are not set up very useful at default, because there will be some extra space above the caption but no space between the caption and the figure or table itself. (Remember: `\belowcaptionskip` is usually set to 0pt.)

Please compare the spacing in these small tables:

Table 1: A table		A	B
		C	D
		A	B
		C	D

Table 2: A table

But you can fix this by using the option `position=`: It specifies how the spacing above and below the caption will be used:

```
position=top (or position=above)
```

tells the caption package to use the spacing useful for caption *above* the figure or table and

```
position=bottom (or position=below)
```

tells the caption package to use the spacing useful for captions *below* the figure or table. (The last one is the default setting except for `longtables`.)

So adding an extra `\captionsetup{position=top}` to the left example table gives you proper spacing around both captions:

Table 3: A table

A	B
C	D

A	B
C	D

Table 4: A table

(Technically speaking `\abovecaptionskip` and `\belowcaptionskip` will be swapped if you specify the option `position=top`, so in both cases `\abovecaptionskip` will be used between the caption and the figure or table itself.)

This option is especially useful when used together with the optional argument of the `\captionsetup` command. (See section 4: “*Useful stuff*” for details)

For example

```
\captionsetup[table]{position=top}
```

tableposition=
New feature
v3.0a

causes all captions within tables to be treated as captions *above* the table (regarding spacing around it). Because this is a very common setting the caption package offers an abbreviating option for the use with `\usepackage`:

```
\usepackage[... ,tableposition=top]{caption}2
```

is equivalent to

```
\usepackage[...]{caption}
\captionsetup[table]{position=top}
```

4 Useful stuff

`\caption` The command

```
\caption[⟨lst.entry⟩]{⟨heading⟩}
```

typesets the caption inside a floating environment like `figure` or `table`. Well, you already know this, but the caption package offers an extension: If you leave the argument `⟨lst.entry⟩` empty, no entry in the list of figures or tables will be made. For example:

```
\caption[] {A figure without entry in the list of figures.}
```

`\caption*` The `longtable` package defines the command `\caption*` which typesets the caption without label and without entry in the list of tables. An example:

²Please note that this is *not* sufficient when using a KOMA-Script document class, you need to use the *global* option `tablecaptionabove`, too.

```

\begin{longtable}{cc}
\caption*{A table}\\
A & B \\
C & D \\
\end{longtable}

```

looks like

A table

A	B
C	D

This package offers this feature, too, so you can use this command now within every floating environment like `figure` or `table`, like:

```

\begin{table}
\caption*{A table}
\begin{tabular}{cc}
A & B \\
C & D \\
\end{tabular}
\end{table}

```

`\captionof` Sometimes you want to typeset a caption *outside* a floating environment, putting a figure within a minipage for instance. For this purpose the `caption` package offers the command

```

\captionof{<float type>}[<lst_entry>]{<heading>} .

```

Note that the first argument, the `<float type>`, is mandatory here, because the `\captionof` command needs to know which name to put into the caption label (e.g. “Figure” or “Table”) and in which list to put the contents entry. An example:

```

\captionof{figure}{A figure}
\captionof{table}{A table}

```

typesets captions like this:

Figure 21: A figure

Table 6: A table

The star variant `\captionof*` has the same behaviour as the `\caption*` command: it typesets the caption without label and without entry to the list of figures or tables.

Please use both `\captionof` and `\captionof*` only *inside* environments (like `minipage` or `\parbox`), otherwise a page break can appear between content and caption. Furthermore some strange effects could occur (e.g., wrong spacing around captions).

`\ContinuedFloat` Sometimes you want to split figures or tables without giving them their own reference number. This is what the command

`\ContinuedFloat`

is for; it should be used as first command inside the floating environment. It prevents the increment of the relevant counter (usually done by `\caption`) so a figure or table with a `\ContinuedFloat` in it gets the same reference number as the figure or table before.

An example:

```
\begin{table}
\caption{A table}
...
\end{table}
...
\begin{table}\ContinuedFloat
\caption{A table (cont.)}
...
\end{table}
```

gives the following result:

Table 7: A table

...

Table 7: A table (cont.)

`\captionsetup` We already know the `\captionsetup` command (see section 2: “Using the package”), but this time we get enlighten about its optional argument *float type*.

Remember, the syntax of this command is

```
\captionsetup[float type]{options}
```

If a *float type* gets specified, all the *options* don’t change anything at this time. Instead they only get marked for a later use, when a caption inside of a floating environment of the particular type *float type* gets typeset. For example

```
\captionsetup[figure]{options}
```

forces captions within a `figure` environment to use the given *options*.

Here comes an example to illustrate this:

```
\captionsetup{font=small}
\captionsetup[figure]{labelfont=bf}
```

gives captions like this:

Figure 22: A figure

Table 8: A table

As you see the command `\captionsetup[figure]{labelfont=bf}` only changed the font of the figure caption labels, not touching all other ones.

`\clearcaptionsetup` If you want to get rid of these parameters marked for an automatic use within a particular environment you can use the command

`\clearcaptionsetup{<float type>}` .

For example `\clearcaptionsetup{figure}` would clear the extra handling in the example above:

Figure 23: A figure

Table 9: A table

As *<float type>* you can usually give one of these two only: `figure` or `table`. But as we will see later some L^AT_EX packages (like the `float`, `longtable`, and `sidecap` package for example) offer additional floating environments and these two commands can also be used with them.

5 Do it yourself!

A family of commands is provided to allow users to define their own formats. This enables information on separators, justification, fonts, and styles to be associated with a name and kept in one place (these commands need to appear in the document preamble, this is the part between `\documentclass` and `\begin{document}`).

`\DeclareCaptionFormat`

You can define your own caption formats using the command

`\DeclareCaptionFormat{<name>}{<code using #1, #2 and #3>}` .

At usage the system replaces #1 with the caption label, #2 with the separator and #3 with the text. So the standard format `plain` is pre-defined by the caption package as

`\DeclareCaptionFormat{plain}{#1#2#3\par}`

`\DeclareCaptionLabelFormat`

Likewise you can define your own caption label formats:

`\DeclareCaptionLabelFormat{<name>}{<code using #1 and #2>}`

At usage #1 gets replaced with the name (e.g. “figure”) and #2 gets replaced with the reference number (e.g. “12”).

`\bothIfFirst`
`\bothIfSecond`

If you define your own caption label formats and use the `subfig` package[10], you should take care of empty caption label names. For this purpose the commands

`\bothIfFirst{<first arg>}{<second arg>}` and
`\bothIfSecond{<first arg>}{<second arg>}`

are offered. `\bothIfFirst` tests if the first argument exists (means: is not empty), `\bothIfSecond` tests if the second argument exists. If yes, both arguments get typeset, otherwise none of them.

For example the standard label format `simple` is *not* defined as

`\DeclareCaptionLabelFormat{simple}{#1 #2}` ,

because this could cause an extra space if #1 is empty. Instead `simple` is defined as

`\DeclareCaptionLabelFormat{simple}{\bothIfFirst{#1}{ }#2}`
,

causing the space to appear only if the label name is present.

`\DeclareCaptionLabelSeparator` You can define your own caption label separators with

```
\DeclareCaptionLabelSeparator{<name>}{<code>}
```

Again an easy example taken from the caption package itself:

```
\DeclareCaptionLabelSeparator{colon}{: }
```

`\DeclareCaptionJustification` You can define your own caption justifications with

```
\DeclareCaptionJustification{<name>}{<code>}
```

The `<code>` simply gets typeset just before the caption. E.g. using the justification `raggedright`, which is defined as

```
\DeclareCaptionJustification{raggedright}{\raggedright}
,
```

lets captions with all lines moved to the left margin.

`\DeclareCaptionFont` You can define your own caption fonts with

```
\DeclareCaptionFont{<name>}{<code>}
```

For example this package defines the options `small` and `bf` as

```
\DeclareCaptionFont{small}{\small} and
\DeclareCaptionFont{bf}{\bfseries}
```

New description
v3.0h

The line spacing could be customized using the `setspace` package, for example:

```
\usepackage{setspace}
\captionsetup{font={onehalfspacing,small},labelfont=bf}
```

Figure 24: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

An example which brings color into life:

```
\usepackage{color}
\DeclareCaptionFont{red}{\color{red}}
\DeclareCaptionFont{green}{\color{green}}
\DeclareCaptionFont{blue}{\color{blue}}
\captionsetup{labelfont=blue,textfont=green}
```

Figure 25: White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

`\DeclareCaptionStyle` You can define your own caption styles with

```
\DeclareCaptionStyle{<name>}[<additional options>]{<options>}
```

Remember, caption styles are just a collection of suitable options, saved under a given name. You can wake up these options at any time with the option `style=<style name>`. All caption styles are based on the default set of options. (See section 3.5: “*Styles*” for a complete list.) So you only need to specify options which are different to them.

If you specify *<additional options>* they get used in addition when the caption fits into a single line and this check was not disabled with the option `singlelinecheck=off`.

Again a very easy example taken from the core of this package: The caption style default is pre-defined as

```
\DeclareCaptionStyle{default}[justification=centering]{}
.
```

5.1 Further Examples

If you would like to have a colon *and* a line break as caption separator you could define it this way:

```
\DeclareCaptionLabelSeparator{period-newline}{. \\\}
```

Selecting this separator with `\captionsetup{labelsep=period-newline}` you get captions like this:

Figure 26.

White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe’s finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

For short captions—which fit into one single line—this separator may not be satisfying, even when the automatically centering process is switched off (with `singlelinecheck=off`):

Figure 27.

A figure.

An own caption style which selects another caption separator automatically puts this right:

```
\DeclareCaptionStyle{period-newline}%
[labelsep=period]{labelsep=period-newline}
```

Figure 27. A figure.

If you would like to keep the centering of these captions instead, an appropriate definition would be something like

```
\DeclareCaptionStyle{period-newline}%
[labelsep=period, justification=centering]%
{labelsep=period-newline} .
```

Using this definition short captions look like

Figure 27. A figure.

while long ones still have a line break after the caption label.

Slightly changed, you also get centered captions if they are longer than one line:

```
\DeclareCaptionStyle{period-newline}%
[labelsep=period]%
{labelsep=period-newline, justification=centering}
```

Figure 28.

White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

Another example: You want captions to look like this:

White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

(Figure 29)

You could do it this way:

```
\DeclareCaptionFormat{reverse}{#3#2#1}
\DeclareCaptionLabelFormat{fullparens}{(\bothIfFirst{#1}{ }#2)}
\DeclareCaptionLabelSeparator{fill}{\hfill}
\captionsetup{format=reverse, labelformat=fullparens,
labelsep=fill, font=small, labelfont=it}
```

Another example: The caption text should go into the left margin; a possible solution would be:

```
\DeclareCaptionFormat{llap}{\llap{#1#2}#3\par}
\captionsetup{format=llap, labelsep=quad, singlelinecheck=no}
```

As a result you would get captions like this:

Figure 30 White sand beaches. The pink smoothness of the conch shell. A sea abundant with possibilities. Duty-free shops filled with Europe's finest gifts and perfumes. Play your favorite game of golf amidst the tropical greens on one of the many championship courses.

6 Using non-standard document classes

New description
v3.0p

The caption package was developed using the standard document classes `article`, `report` and `book`, it also works with the Dutch document classes `artikel`, `rapport` and `boek`. It is *incompatible* with all other document classes, except the ones based on one of the above.

If you are unsure if your document class is supported or not, it's better *not* to use this package. You could try if your document will compile fine with the caption package anyway, but you should watch carefully if side-effects occur, i.e. the look and feel of your captions should *not* change if you just include the caption package without options. (If it does, you can carry on, but you should be warned that the output could change with upcoming versions of the caption package.) If this is fine, you can start setting

options with `\usepackage[...]{caption}` or `\captionsetup` and keep your fingers crossed.

The upcoming version v3.1 of the caption package will be adapted to several document classes, e.g. the beamer class and the KOMA-Script ones. It's planned for release during Summer 2007.

7 Compatibility to other packages

The caption package contains special adaptations to other packages, so the captions should always look like you have specified them to look like.

These are the packages the caption package is adapted to:

<code>float</code>	Gives you the possibility to define new floating environments
<code>listings</code>	Typesets source code listings
<code>longtable</code>	Typesets tables spanned over multiple pages
<code>rotating</code>	Supports rotated figures and tables
<code>sidecap</code>	Offers captions <i>beside</i> figures or tables
<code>supertabular</code>	Typesets tables spanned over multiple pages

New feature
v3.0b

If you use one of the above packages together with the caption package you get the additional possibility to set up captions with

```
\captionsetup[<environment>]{<options>}
```

where *<environment>* stands for any environment the above packages offer. (Please note that this do not work with the `sideways` environments offered by the rotating package.) For example

```
\captionsetup[lstlisting]{labelfont=bf}
```

forces captions inside the `lstlisting` environment to have bold labels.

If a certain support is not desired you can switch it off using the caption package option

```
\usepackage[... ,<package>=no]{caption}
```

For example specifying the option `float=no` means you don't like the caption package to support the float package. (Note: You can specify these options only within the `\usepackage` command, especially *not* at a later time with `\captionsetup`.)

For further information about the packages mentioned above please take a look at the documentation belonging to them or buy yourself The L^AT_EX Companion[1].

7.1 The float package

A very useful feature is provided by the float package[2]: It offers the float placement specifier `H` which is much more restrictive than the specifier `h` offered by L^AT_EX. While the latter one is only a recommendation to L^AT_EX to set the float "here", the `H` forces the float to appear exactly at the spot where it occurs in your input file and nowhere else.

Furthermore it offers different styles for floating environments, these styles are `plain`, `plaintop`, `ruled`, and `boxed`. You can link one of these styles to either new floating environments or to one of the existing environments `figure` and `table`.

If you are using the caption package together with the float package a caption style called `ruled` gets defined automatically:

```
\DeclareCaptionStyle{ruled}{labelfont=bf,labelsep=space}
```

This style represents the caption layout in `ruled` styled floats. For you as an end user this means that captions within `ruled` floats will always look like this, nevertheless what generic caption options do you specify:

Program 7.1 The first program. This hasn't got anything to do with the package but is included as an example. Note the `ruled` float style.

```
#include <stdio.h>

int main(int argc, char **argv)
{
    for (int i = 0; i < argc; ++i)
        printf("argv[%d] = %s\n", i, argv[i]);
    return 0;
}
```

If you want a different layout for `ruled` captions you have to define your own one using the command

```
\DeclareCaptionStyle{ruled}{{options}}
```

This mechanism also works with all other float styles. If you want a special caption layout—for `plain` or `boxed` floats for example—you can simply define a suitable caption style with the same name as the float style.

Note: For successful cooperation you need the float package version 1.3 or newer.

7.2 The listings package

New description
v3.0b

The listings package[6] is a source code printer for $\text{\LaTeX}$. You can typeset stand alone files as well as listings with an environment similar to `verbatim` as well as you can print code snippets using a command similar to `\verb`. Many parameters control the output and if your preferred programming language isn't already supported, you can make your own definition.

Note: For successful cooperation you need the listings package version 1.2 or higher. You'll get an error message when using an older version!

7.3 The longtable package

The longtable package[7] offers the environment `longtable` which behaves similar to the `tabular` environment, but the table itself can span multiple pages.

Note: For successful cooperation you need the longtable package version 3.15 or newer.

7.4 The rotating package

The rotating package[8] offers the floating environments `sidewaysfigure` and `sideways-table` which are just like normal figures and tables but rotated by 90 degree. Furthermore they always use a full page on their own.

7.5 The sidecap package

New description
v3.0b

The sidecap package[9] offers the floating environments `SCfigure` and `SCtable` which are like normal figures and tables but the caption will be put *beside* the contents.

The sidecap package offers it's own options for justification. If set, they will override the one specified with the caption option `justification=` for captions beside their contents.

`listof=` Using the sidecap package you will probably notice that suppressing the entry in the list of figures or tables with `\caption[]{\dots}` won't work inside these environments. This is caused by the implementation design of the sidecap package, but you can use `\captionsetup{listof=false}` inside the figure or table as an alternative here.

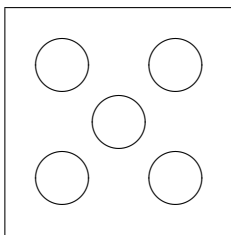


Figure 31: A small example with the caption beside the figure.

7.6 The supertabular package

The supertabular package[11] offers the environment `supertabular` which is quite similar to the `longtable` environment provided by the `longtable` package. Both offers the typesetting of tables which can span multiple pages. For a detailed discussion about the differences between these powerful packages please take a look at The L^AT_EX Companion[1].

7.7 Known incompatibilities

New description
v3.0b

Using the caption package together with one of the following packages is not recommended; usually this would cause unwanted side effects or even errors:

`ccaption`, `ftcap`, `hvfloating`, and `nonfloat`

8 Compatibility to older versions

8.1 The caption package version 1.x

This version of the caption package still supports the old options and commands provided by the version 1.x of this package. So there shouldn't occur any problems compiling old

documents, but please don't mix old options and commands with the new ones. This isn't supported and can cause ugly side effects.

Here comes a short oversight of the obsolete options and commands and how they have been replaced within this version of the caption package:

caption v1.x	caption v3.x
normal	format=plain
hang	format=hang
isu	format=hang
center	justification=centering
centerlast	justification=centerlast
nooneline	singlelinecheck=off
scriptsize	font=scriptsize
footnotesize	font=footnotesize
small	font=small
normalsize	font=normalsize
large	font=large
Large	font=Large
up	labelfont=up
it	labelfont=it
sl	labelfont=sl
sc	labelfont=sc
md	labelfont=md
bf	labelfont=bf
rm	labelfont=rm
sf	labelfont=sf
tt	labelfont=tt

Beside the options for setting up the desired font there were also the commands `\captionsize` resp. `\captionfont` and `\captionlabelfont` who could be redefined with `\renewcommand` and allowed an alternate and more flexible way to change the font used for captions. This mechanism was replaced by the commands

```
\DeclareCaptionFont{...}{...}    and
\captionsetup{font=...,labelfont=...} .
```

(See section 5: “*Do it yourself*”)

Setting the margin for captions was done in v1.x with

```
\setlength{\captionmargin}{...} .
```

This was replaced by

```
\captionsetup{margin=...} .
```

(See section 3.4: “*Margins and further paragraph options*”)

For example the old-style code

```
\usepackage[hang,bf]{caption}
\renewcommand\captionfont{\small\sffamily}
\setlength\captionmargin{10pt}
```

will still work fine, but should be written today as

```
\usepackage[format=hang,labelfont=bf,font={small,sf},
margin=10pt]{caption}
```

or

```
\usepackage{caption}
\captionsetup{format=hang,labelfont=bf,font={small,sf},
margin=10pt} .
```

The quite exotic option `ruled` which allowed a partial usage of the caption settings for ruled floats defined with the `float` package will be emulated by this version of the caption package, too. But using this option is not recommended anymore since this version of the caption package offers a more flexible way for changing the captions of these floating environments:

```
\DeclareCaptionStyle{ruled}{...}
```

resp.

```
\captionsetup[ruled]{...} .
```

(See section 5: “*Do it yourself*”, 4: “*Useful stuff*”, and 7.1: “*The float package*”)

8.2 The caption2 package version 2.x

Although they do very similar stuff, the packages `caption` and its experimental and now obsolete variant `caption2` have a very different implementation design. Therefore a full compatibility could not be offered. For that reason you will still find a file called `caption2.sty` in this package distribution, so old documents using the `caption2` package will still compile fine.

Newly created documents should use the actual version of the `caption` package instead. In most cases it's sufficient to replace the command

```
\usepackage[...]{caption2}
```

by

```
\usepackage[...]{caption} .
```

But some options and commands will not be emulated, so you can get error messages afterwards. This section will hopefully help you removing these errors. If you have problems migrating from `caption2` to `caption` please don't hesitate to send me an e-mail asking for help.

In addition to the obsolete options shown in the last section these ones will be emulated, too:

caption2 v2.x	caption v3.x
<code>flushleft</code>	<code>justification=raggedright</code>
<code>flushright</code>	<code>justification=raggedleft</code>
<code>oneline</code>	<code>singlelinecheck=on</code>

Setting the margin for captions was done in v2.x with

```
\setcaptionmargin{...} resp. \setcaptionwidth{...} .
```

This was replaced by

```
\captionsetup{margin=...} resp. \captionsetup{width=...} .
```

(See section 3.4: “*Margins and further paragraph options*”)

Setting an indentation was done in v2.x with

```
\captionstyle{indent}  
\setlength\captionindent{...} .
```

This is now done with

```
\captionsetup{format=plain,indentation=...} .
```

The so-called single-line-check was controlled by the commands `\onelinecaptionsfalse` (for switching the check off) and `\onelinecaptionstrue` (for switching the check on). This was replaced by `\captionsetup{singlelinecheck=off}` resp. `\captionsetup{singlelinecheck=on}`. (See section 3.2: “*Justification*”)

The commands

```
\captionlabeldelim, \captionlabelsep, \captionstyle,  
\defcaptionstyle, \newcaptionstyle, and \renewcaptionstyle
```

do not have a simple replacement and therefore will not be emulated by this version of the caption package. (So using them will cause error messages.) Rewriting such code is not always easy and straight-ahead, but by conscientious reading of this manual you should find appropriate options and commands instead.

The v2.x option `ignoreLTcapwidth` do not have a replacement, too. But in most cases you can simply drop using that option because in this version of the caption package the value of `\LTcapwidth` will be ignored anyway (unless you set it to a different value than the default one). (See section 7.3: “*The longtable package*”)

9 Further reading

I recommend the following documents for further reading:

- The T_EX FAQ - Frequently asked questions about T_EX and L^AT_EX:
<http://faq.tug.org/>
- A French FAQ can be found at
<http://www.grappa.univ-lille3.fr/FAQ-LaTeX/>
- epslatex from Keith Reckdahl contains many tips around including graphics in L^AT_EX 2_ε documents. You will find this document in the directory
<ftp://ftp.ctan.org/pub/tex/info/epslatex/>

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References

- [1] Frank Mittelbach and Michel Goossens: *The L^AT_EX Companion (2nd. Ed.)*, Addison-Wesley, 2004.
- [2] Anselm Lingnau: *An Improved Environment for Floats*, 2001/11/08
- [3] Olga Lapko: *The floatrow package documentation*, 2005/05/22
- [4] Sebastian Rahtz & Heiko Oberdiek: *Hypertext marks in L^AT_EX*, 2007/01/25
- [5] Heiko Oberdiek: *The hypcap package – Adjusting anchors of captions* 2007/02/19
- [6] Carsten Heinz: *The Listings Package*, 2004/02/13
- [7] David Carlisle: *The longtable package*, 2000/10/22
- [8] Sebastian Rahtz and Leonor Barroca: *A style option for rotated objects in L^AT_EX*, 1997/09/26
- [9] Rolf Niepraschk und Hubert Gäßlein: *The sidecap package*, 2003/06/06
- [10] Steven D. Cochran: *The subfig package*, 2005/07/05

- [11] Johannes Braams und Theo Jurriens: *The supertabular environment*, 2002/07/19
- [12] Anne Brüggemann-Klein: *Einführung in die Dokumentverarbeitung*, B.G. Teubner, Stuttgart, 1989